



INDIAN INSTITUTE OF TECHNOLOGY, JODHPUR

HR Analytics & Employee Attrition Analysis Case Study Report

Dataset: IBM HR Analytics Employee Attrition & Performance
Tool: Python (pandas, seaborn, matplotlib) | Records: 1,470 | Variables: 35

COLAB LINK : https://colab.research.google.com/drive/1NUns8m9N74quoX0HZ-jZOiXyq3bZ-L7m#scrollTo=x92s56VC_cMy

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Course Visualization and Storytelling with Data

1. OBJECTIVE

This case study investigates employee attrition within an organization using the IBM HR Analytics dataset. The primary objective is to identify the key drivers of voluntary resignation, determine which employee segments are most affected, and derive actionable recommendations that can support evidence-based HR decision-making. The analysis covers departmental attrition patterns, the influence of overtime on turnover, salary and retention relationships, satisfaction metrics, and demographic experience trends.

2. DATASET OVERVIEW

Attribute	Detail
Source	IBM HR Analytics — WA_Fn-UseC_-HR-Employee-Attrition.csv
Total Records	1,470 employee records
Total Features	35 variables (numeric & categorical)
Target Variable	Attrition (Yes / No)
Missing Values	None — dataset is complete
Data Type	Cross-sectional, HR administrative records

3. KEY PERFORMANCE INDICATORS

Total Employees	Attrition Count	Attrition Rate	Average Age	Avg. Monthly Income
1,470	237	16.12%	36.92 yrs	\$6,503

The organization employs 1,470 individuals across three departments. Of these, 237 employees have exited, yielding a total attrition rate of 16.12%. The average employee age stands at 36.92 years, with an average monthly income of \$6,503. The attrition rate exceeds the industry benchmark of 10–12%, indicating a meaningful retention challenge that warrants structured investigation.

4. ATTRITION OVERVIEW

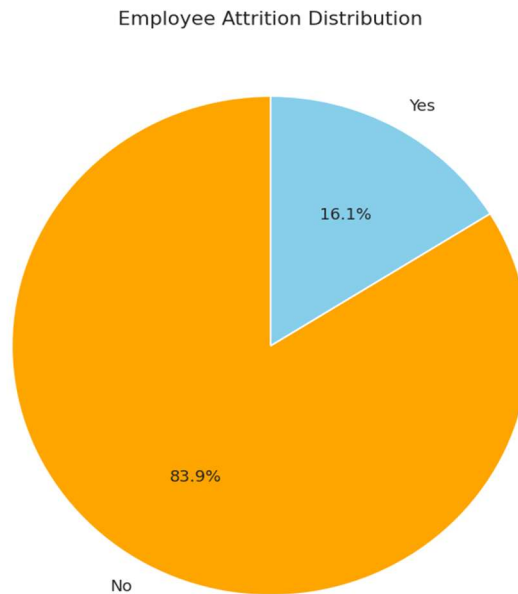


Figure: Employee Attrition Distribution — 16.1% Yes vs. 83.9% No

The attrition distribution reveals that approximately 1 in 6 employees has left the organization. While the majority of the workforce (83.9%) remains, the 16.12% attrition rate translates to significant replacement and training costs. Understanding the root causes is essential for building effective retention strategies.

5. WORKFORCE & DEMOGRAPHIC OVERVIEW

5.1 Employee Age Distribution

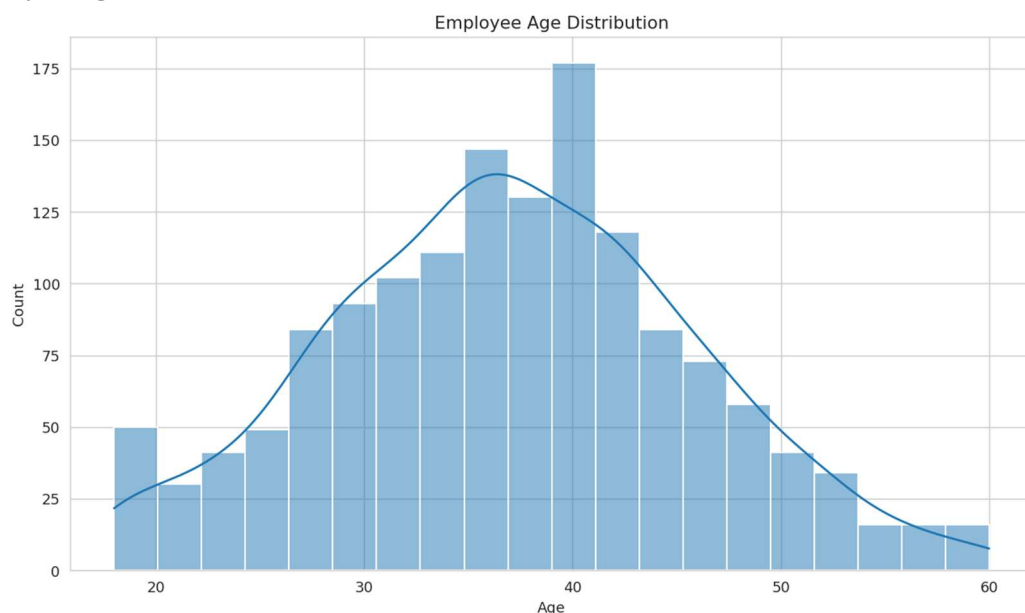


Figure: Employee Age Distribution with KDE Curve

The workforce age distribution is approximately normal, centered around 36–37 years. A notable cluster of employees falls between ages 28–40, representing the most active working-age group. Younger employees (under 30) form a significant segment and, based on attrition patterns observed later, show higher propensity to resign. The tail extending beyond 50 represents senior staff with typically lower attrition rates.

5.2 Gender Distribution

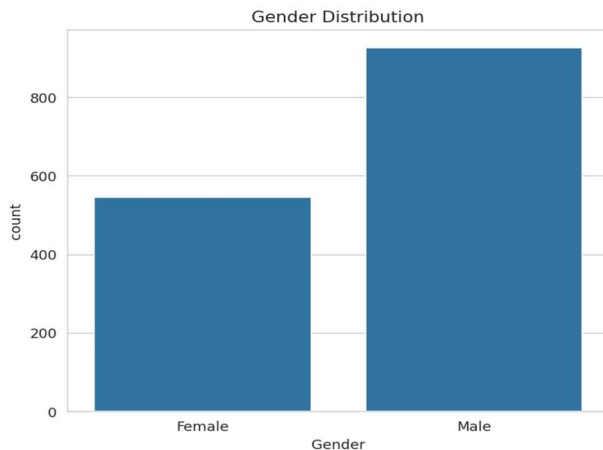


Figure: Gender Distribution

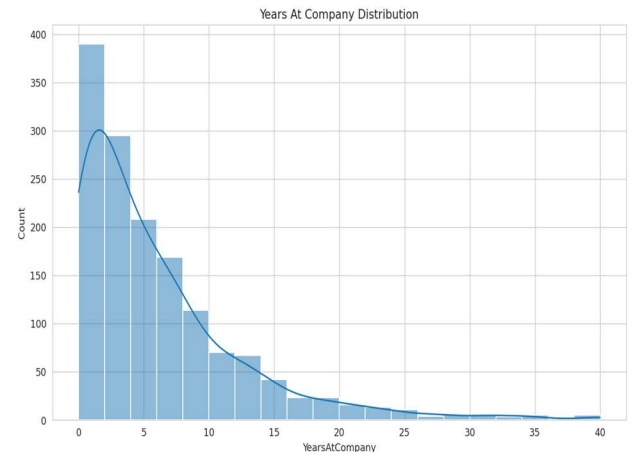


Figure: Years at Company Distribution

Males constitute approximately 60% of the workforce while females account for 40%. The gender gap is moderate but consistent across departments. The tenure distribution (Years at Company) is right-skewed, with a large concentration of employees having 0–5 years of experience in the current role, indicating a relatively younger workforce in terms of organizational tenure.

6. DEPARTMENT & JOB ROLE ANALYSIS

6.1 Department-wise Attrition

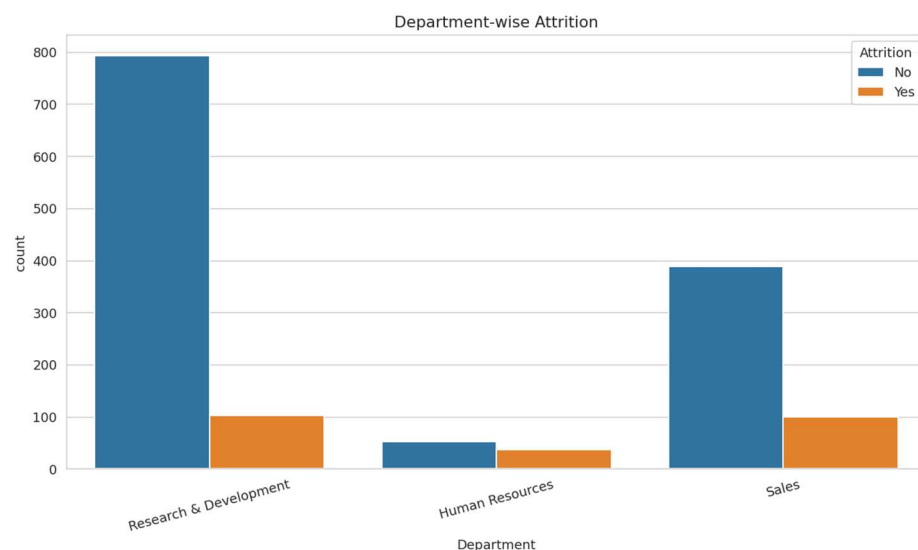


Figure: Department-wise Employee Attrition — Sales, R&D, and Human Resources

Research & Development holds the largest employee share and consequently records the highest absolute attrition count. However, the Sales department exhibits a disproportionately high attrition rate relative to its headcount, suggesting that role pressure, commission-based pay structures, and targets may be contributing to turnover. Human Resources, while the smallest department, shows the highest attrition rate in percentage terms, pointing to operational stress and limited career mobility within the function.

6.2 Job Role vs. Attrition

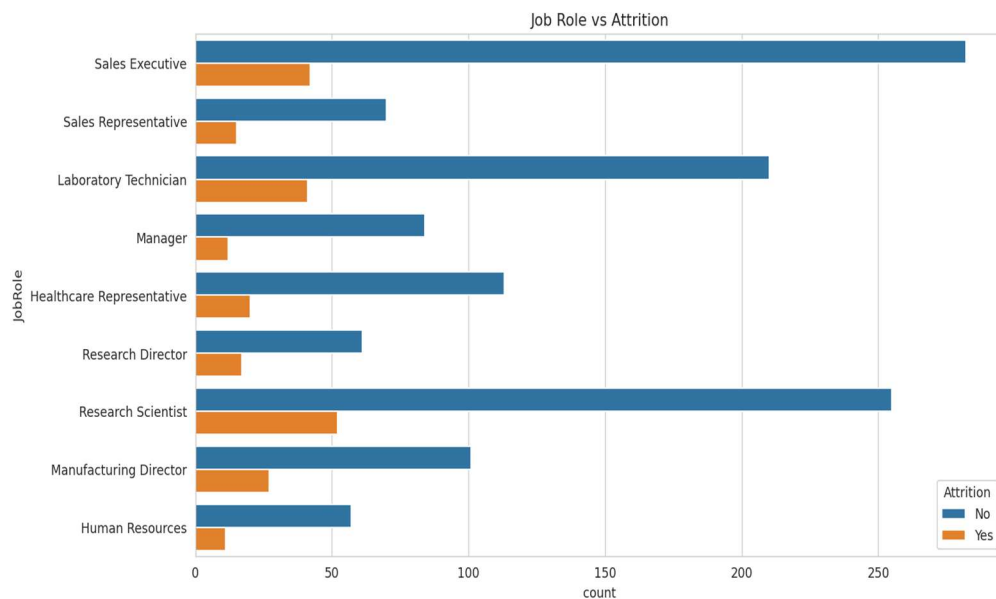


Figure: Job Role vs. Attrition — All Nine Job Roles Compared

Laboratory Technicians, Sales Executives, and Research Scientists record the highest attrition in absolute numbers. Sales Representatives, despite smaller headcount, show a high proportional attrition rate. Senior roles such as Manager and Research Director reflect low attrition, consistent with findings that job stability and compensation tend to improve significantly at higher levels. The pattern indicates that frontline and junior roles carry the heaviest attrition burden.

7. ATTRITION DRIVER ANALYSIS

7.1 Overtime & Attrition

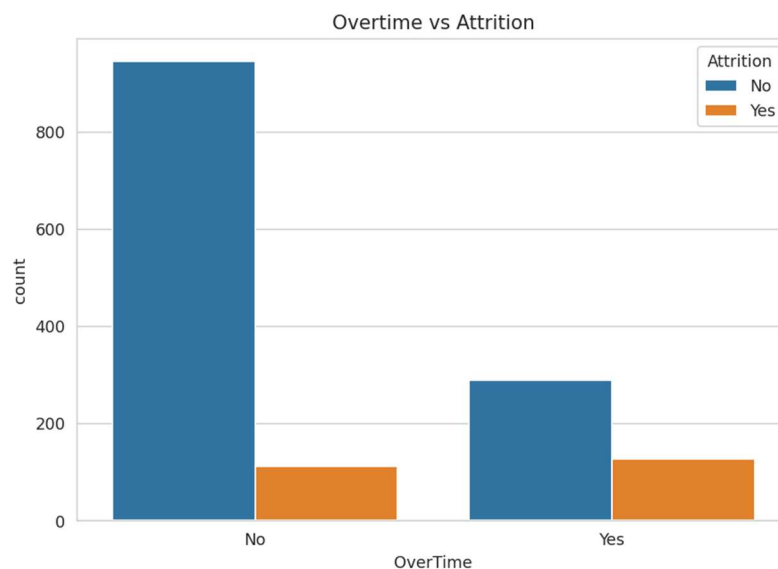


Figure: Overtime vs. Attrition — Employees Required to Work Overtime

Overtime is among the strongest individual predictors of attrition in this dataset. Employees required to work overtime show a substantially higher rate of resignation compared to those who do not. Among attrited employees, a majority were working overtime, suggesting that chronic overwork, inadequate work-life boundary management, and burnout are primary contributors to voluntary exit. This finding has immediate policy implications for scheduling, workload distribution, and compensation for extended hours.

7.2 Monthly Income Distribution

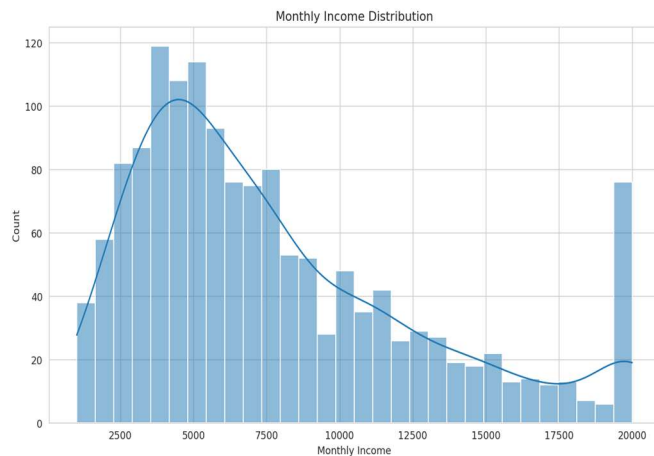


Figure: Monthly Income Distribution

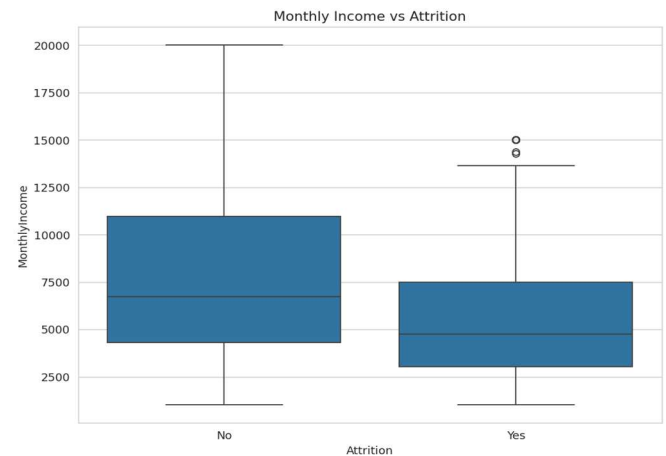


Figure: Monthly Income vs. Attrition (Boxplot)

The income distribution is right-skewed, with the majority of employees earning between \$2,000 and \$8,000 per month. The boxplot comparison reveals that employees who left the organization (Attrition = Yes) had a noticeably lower median monthly income than those who stayed. The interquartile range for attrited employees is compressed toward the lower income bands, confirming that compensation below market expectations is a meaningful driver of resignation decisions.

7.3 Experience vs. Salary

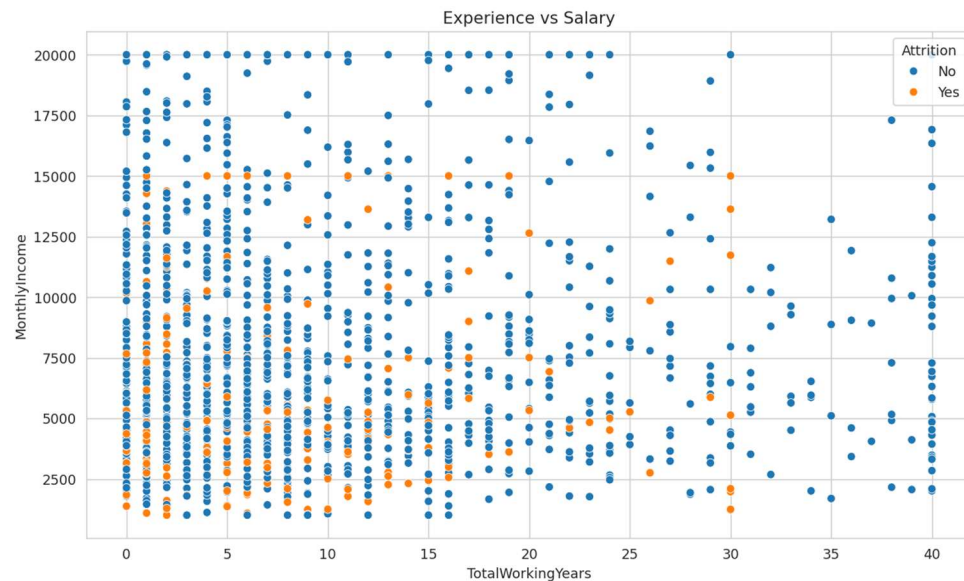


Figure: Total Working Years vs. Monthly Income — Colored by Attrition

The scatter plot illustrates a positive relationship between total working experience and monthly income, as expected. Attrited employees (highlighted) are concentrated in the lower-left cluster — representing younger, less experienced employees earning below-average salaries. Few high-experience, high-income employees are observed in the attrition group, reinforcing that early-career employees in lower pay bands represent the highest retention risk. Targeted compensation reviews at the junior and mid-level bands could materially reduce turnover.

7.4 Education Field vs. Attrition

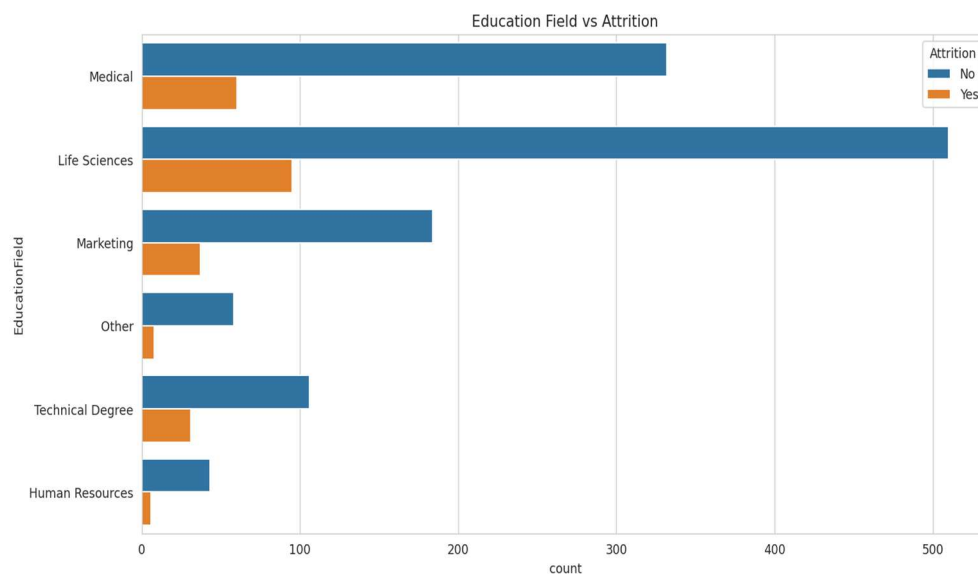


Figure: Education Field vs. Attrition — All Education Backgrounds Compared

Life Sciences and Medical education backgrounds dominate the workforce and also account for the largest attrition counts in absolute terms. Marketing graduates, though fewer in number, show a notable proportion of attrition. Technical Degree holders demonstrate relatively lower attrition, possibly due to specialized skill premiums. Human Resources graduates show the smallest absolute numbers. These patterns suggest that role-to-education alignment and career pathing within specific educational cohorts require closer management attention.

8. SATISFACTION & WELL-BEING ANALYSIS

8.1 Job Satisfaction vs. Attrition

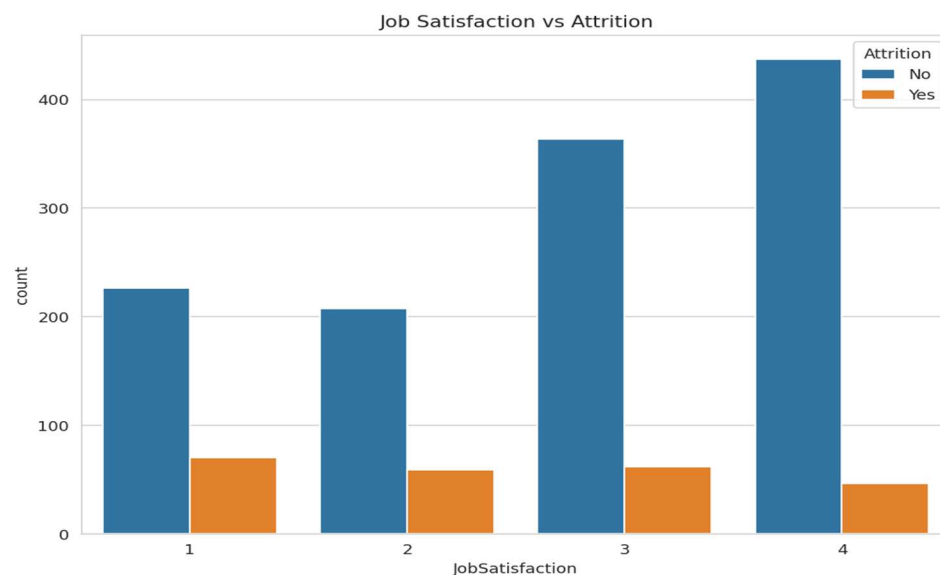


Figure: Job Satisfaction (Scale 1–4) vs. Attrition — Lower Scores Linked to Higher Exit

Job satisfaction was measured on a 1-to-4 scale (1 = Low, 4 = High). The countplot reveals that employees with satisfaction scores of 1 and 2 exhibit a proportionally higher share of attrition compared to those rating 3 or 4. At satisfaction level 1, the proportion of exiting employees is markedly elevated, indicating that dissatisfaction is a direct precursor to resignation. Employees reporting the highest satisfaction (level 4) demonstrate the strongest retention, underscoring the return on investment in engagement programs.

8.2 Work-Life Balance vs. Attrition

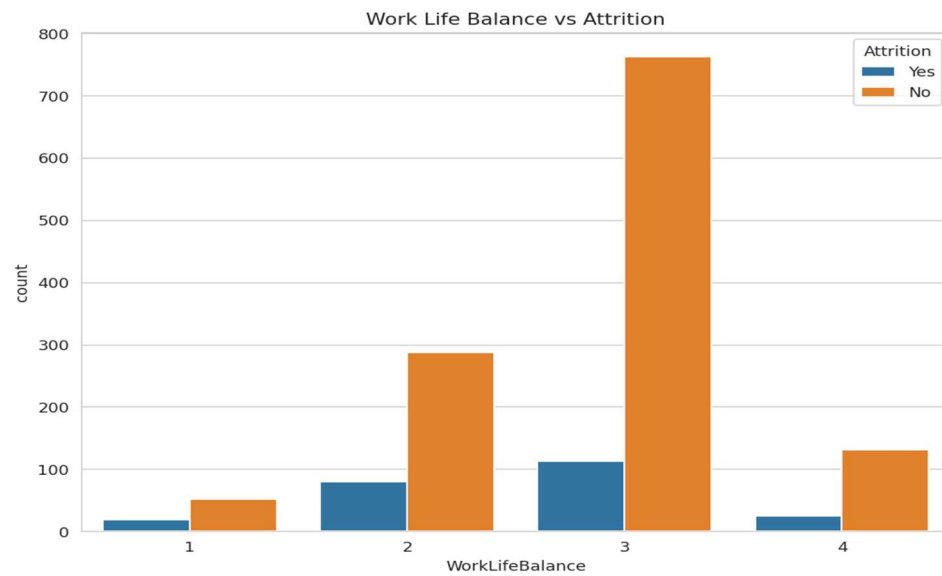


Figure: Work-Life Balance (Scale 1–4) vs. Attrition — Poor Balance Drives Higher Turnover

Work-life balance was rated on a 1-to-4 scale (1 = Bad, 4 = Best). The data clearly shows that employees at balance level 1 have the highest relative attrition proportion. Balance levels 2 and 3 constitute the majority of the workforce, with attrition distributed across both groups. Very few employees rate their work-life balance at level 4, but those who do show minimal attrition. Combined with the overtime finding, these results confirm that workload manageability and personal time preservation are critical retention levers.

8.3 Correlation Analysis

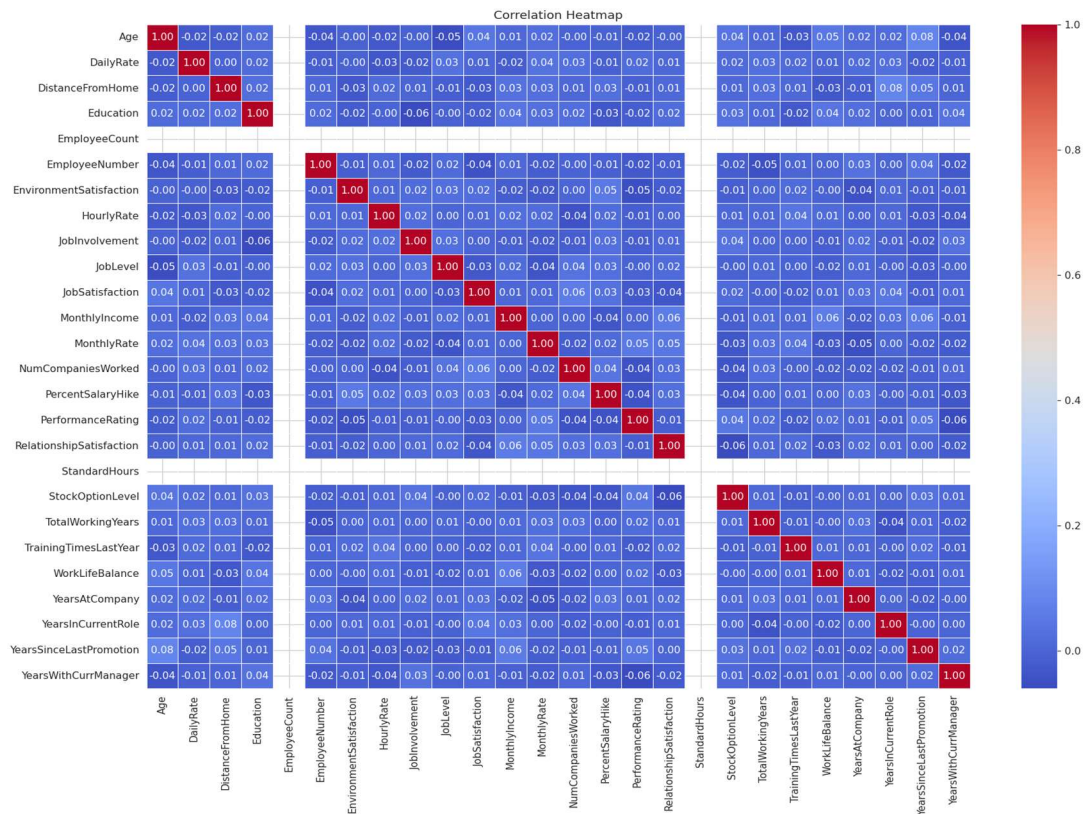


Figure: Correlation Heatmap — Numeric Features (coolwarm palette)

The correlation heatmap reveals several key structural relationships. Monthly income is positively correlated with job level, total working years, and years at company — confirming that experience and seniority drive compensation. Age correlates moderately with total working years and years at company, as expected. Job level shows a strong positive correlation with monthly income and a moderate relationship with years in the current role. Notably, daily rate, hourly rate, and monthly rate show weak correlations with most variables, indicating they are not reliable standalone predictors of attrition. The heatmap guides feature selection for any predictive modeling extensions of this study.

9. KEY BUSINESS INSIGHTS

- Employees working overtime exhibit significantly higher attrition rates, pointing to chronic workload pressure as a primary exit trigger.
- The Sales department and Sales Executive role carry disproportionately high attrition relative to workforce share.
- Lower-income employees are far more likely to resign, with attrited employees concentrated in the below-average salary bracket.
- Job satisfaction levels of 1 and 2 are directly associated with elevated attrition, confirming engagement as a measurable retention variable.
- Poor work-life balance (rating 1) correlates strongly with employee departure, amplifying the overtime effect.
- Younger employees with shorter tenure and fewer total working years form the highest-risk attrition cohort.
- Human Resources, despite being the smallest department, shows the highest proportional attrition rate among all departments.
- Laboratory Technicians and Research Scientists account for the largest absolute attrition counts within the R&D department.

10. KEY FINDINGS

1. The overall attrition rate of 16.12% exceeds standard industry benchmarks (10–12%), confirming a systemic retention challenge rather than isolated incidents.
2. Overtime obligation is the single most visually prominent binary driver of attrition. Its removal or compensation reform would likely yield the fastest measurable reduction in turnover.
3. Compensation gaps at junior and mid-level bands are a structural vulnerability. Early-career employees earning below-average incomes are the highest attrition risk group.
4. Job satisfaction and work-life balance scores function as early-warning indicators. Monitoring these through regular pulse surveys can enable preemptive intervention.
5. Department-specific attrition patterns suggest that managerial culture, role demands, and career visibility differ significantly across Sales, R&D, and Human Resources.
6. Experience and salary move together; however, high-experience employees rarely appear in the attrition group, confirming that long-term retention is achievable with appropriate career and pay progression.

11. RECOMMENDATIONS

	Recommendation	Expected Outcome
1	Introduce overtime caps and compensatory pay policies for roles exceeding standard hours	Reduce attrition among overtime-affected employees; lower burnout rates
2	Conduct salary benchmarking for junior and mid-level roles against industry standards	Increase competitiveness of compensation; reduce salary-driven exits
3	Deploy quarterly employee engagement and job satisfaction surveys	Enable early identification of dissatisfied employees before resignation decision is made
4	Design structured career development pathways for Sales Executives and Lab Technicians	Improve perceived growth opportunities; reduce early-career attrition
5	Establish a work-life balance monitoring framework with managerial accountability	Improve retention among employees scoring 1–2 on work-life balance scale
6	Implement targeted retention programs for employees in 0–5 year tenure bracket	Stabilize the highest-risk attrition cohort; reduce onboarding and replacement costs
7	Review Human Resources department structure, workload, and career mobility options	Address disproportionately high attrition rate within the smallest department

12. CONCLUSION

This case study presents a structured, data-driven investigation into employee attrition at a large organization using the IBM HR Analytics dataset. The analysis confirms that attrition is not random but follows identifiable patterns linked to overtime demands, compensation levels, job satisfaction, work-life balance, and career experience. The 16.12% attrition rate reflects organizational friction that, left unaddressed, carries measurable financial and operational costs in the form of recruitment, onboarding, and productivity loss.

The evidence consistently points to a cluster of early-career, lower-compensated, overtime-exposed employees as the primary attrition risk segment. Addressing this segment through compensation reform, structured career development, and workload management would deliver the most immediate retention impact. Simultaneously, ongoing measurement of satisfaction and work-life balance provides the organizational intelligence necessary for proactive HR management.

The visualizations produced through Python-based exploratory data analysis serve as the analytical foundation of this report. Each chart directly supports the findings presented, ensuring that all conclusions are grounded in observed data patterns rather than assumptions. This report provides a replicable template for similar HR analytics investigations and can be extended with predictive modeling to build a real-time attrition risk scoring system.

Thank you

Dataset: WA_Fn-UseC_-HR-Employee-Attrition.csv | Analysis Tool: Python 3 (pandas, seaborn, matplotlib)